

DEEPAK KUMAR PRASAD

✉ deepak.prasad.ai@gmail.com — [in linkedin.com/in/dpkpr1](https://www.linkedin.com/in/dpkpr1) — [beingdpkpr.github.io/deepak-prasad/](https://github.com/beingdpkpr/deepak-prasad/)

☎ +91 8861 327919 — 📱 +1 (707) 733-3727

📍 Bangalore, India (Open to relocation)

SUMMARY

Software Architect with 9+ years designing large-scale supply chain planning platforms serving 150+ global clients. Deep expertise in distributed systems, solver integrations, multi-layer configuration frameworks, and performance engineering—delivering 50% compute reduction and 30% memory optimization in production environments. Proven ability to define technical direction, lead 10+ person engineering teams, and translate complex domain problems into scalable platform architecture.

EXPERIENCE

o9 Solutions, Bangalore

Senior Application Architect

Application Architect

Senior Software Engineer

Python Developer (Contract)

Feb 2019 – Present

Apr 2025 – Present

Apr 2021 – Apr 2025

Aug 2019 – Apr 2021

Feb 2019 – Aug 2019

- **Architected the Supply Planning platform configuration framework**—a multi-layer system (Product Default, Product Override, Project Config, Tenant Config) supporting namespace-partitioned overrides, hot-reload via atomic replacement, and schema-validated merge precedence. Eliminated config drift across 150+ client deployments and enabled zero-downtime tenant onboarding.
- **Re-engineered core Supply Planning compute pipeline** (TLB, MRP, IOP) with advanced parameterization and memory-efficient data structures, achieving **50% reduction in batch computation time** and **30% decrease in memory usage** across multiple production environments.
- **Designed and integrated 6+ third-party optimization solvers** with the o9 platform—including Samsung NSR solver using Boost serialization and Parquet-based data exchange—standardizing the solver integration contract and improving deployment scalability across client environments.
- **Designed foundational Demand Netting and Order Promising plugins**—now mission-critical modules in production for 150+ global clients—with extensible configuration hooks and observability instrumentation for enterprise-scale deployments.
- **Built an internal developer productivity platform** (Tenant Extractor, Builder, Performance Analyzer, Test Recorder/Runner, Rule Creator, Dataset Multiplier) adopted by 100+ internal teams, improving consultant and developer productivity by **25%** and reducing environment setup time significantly.
- **Established a modular plugin versioning system** managing 30+ plugin variants, enabling independent deployment and customization per client without disrupting the core platform—reducing release risk and accelerating client onboarding cycles.
- **Built GenAI agents for supply planning workflows**—including a Demand Netting & Demand Prioritization agent that explains functionality, customizes workflows to business requirements, and debugs implementation issues for consultants—plus internal productivity agents for the engineering team, integrated with the o9 platform via MCP tools.
- **Built and led engineering teams over 5 years (Apr 2021 – Present)**—starting with a small group of direct reports and scaling to a team of 10+ developers over the last 3 years: defined technical roadmap, conducted architecture reviews, established coding standards, drove performance benchmarking culture, and mentored engineers on supply planning domain and distributed systems design.

- **Architected a workflow orchestration microservice** on Netflix Conductor, customizing the open-source framework for enterprise orchestration requirements and integrating it into production infrastructure.
- **Modernized a large-scale enterprise security platform** by integrating 5+ Google APIs and refactoring core subsystems, improving system reliability and reducing technical debt across the codebase.
- **Built Attendance & Visitor Management systems** (Android) automating check-ins and visitor tracking, reducing manual overhead by **70%** with actionable facility analytics.

EDUCATION

Bachelor of Engineering, Computer Science

Aug 2012 – May 2016

SJCE, Mysore

SKILLS

Technical Skills	
Languages	Python, C++, Java, SQL
Architecture	Distributed Systems, Microservices, Multi-layer Config Frameworks, API Design, Performance Optimization, Solver Integration
Frameworks & Tools	FastAPI, Boost (C++), ReactJS, Streamlit, LangGraph
Data & Messaging	PostgreSQL, Parquet, ETL Pipelines, Data Serialization
AI / Automation	GenAI Agent Development (LangGraph, Claude API, MCP tool integration), Domain-explainer & Debugging Agents, Python automation tooling
DevOps	Docker, Kubernetes, Git, CI/CD, Linux
Domain	Supply Chain Planning (MRP, TLB, IOP, Demand Netting, Order Promising), Technical Leadership

CERTIFICATIONS

Design for Performance Architect — o9 Solutions · Credly	Oct 2023
Claude AI Development (Specialization, 3 courses) — Anthropic Education	Apr 2026
Python for Everybody (Specialization, 5 courses) — University of Michigan · Coursera	May 2017
Introduction to Data Science (Specialization, 4 courses) — IBM · Coursera	Jan 2021
Programming Foundations with JavaScript, HTML and CSS — Duke University · Coursera	Dec 2017

Full portfolio: beingdpkpr.github.io/deepak-prasad/